

Supply Shocks, Demand Shocks and Yield Curve Dynamics ^{*}

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Abstract

This paper presents a framework for extracting the supply and demand shocks using the revisions of expectations obtained from a Survey of Professional Forecasters. The main contribution of this paper is the identification of these shocks and the subsequent assessment of their impact on yield curve dynamics. This allows for the asymmetry in the bond yield movements, where we show that yields react differently to supply and demand shocks. The analysis is performed on the US data ranging from 1969 (q4) to 2022 (q1).

Keywords: Expectation Revision, Supply and Demand Shocks, Yield Curve
JEL: D84, E31, E32, E43

1. Introduction

The practical thinking of many analysts is still dominated by the IS-LM and AD-AS framework and the associated thinking about the business cycle in terms of supply and demand shocks. This reasoning is somewhat diminished in the current academic literature. Nevertheless, also most of the New Keynesian model, which is a workhorse of modern macroeconomics,

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can be interpreted similarly as these models can be expressed in the most straightforward concept using only three equations, e.g., canonical three-equation New Keynesian model. The main difference is the treatment of expectation, which substantially impacts the model solution, but the supply-demand interpretation remains standard.

This paper returns to basics and examines supply and demand shocks and their impact on financial variables, specifically bond yields and Fed funds rate. In terms of empirical work, our study follows many other papers examining the impact of the macroeconomic variables on yield curve dynamics (Joslin et al. (2014), Bauer and Rudebusch (2017)). The fundamental difference compared to many other studies is that we try to identify supply and demand shocks directly and, therefore, distinguish their impact on these financial variables. In this regard, most similar to this paper is a study by Bekaert et al. (2021), which identifies demand and supply shocks based on higher-order moments in the data. For a more theoretical treatment of this topic, see Michailat and Saez (2022). Hence, our research basically offers a different view of a time-varying term premium and natural rate of interest, where we seek a more structural perspective on the given subject, as the observed variability in the bond yields documented by many other studies may be due to different shocks affecting the economy rather than a structural change of the term premium or equilibrium interest rate. For this reason, our paper have close connection to the studies that identified the monetary policy rule (Taylor (1993), Clarida et al. (2000), Orphanides and Williams (2004), Coibion and Gorodnichenko (2011), Carvalho et al. (2021), Bauer et al. (2022)) or evolution of the term premium (Adrian et al. (2013), Bauer and Rudebusch (2020)). Accordingly, our paper also connects to studies that deal with the importance of asymmetries in the economy (Conrad et al. (2013), Baumeister and Hamilton (2018), Bauer and Chernov (2021)).

In terms of methodology, we freely follow studies dealing with estimating monetary policy surprise. These studies try to isolate this shock based on high-frequency data and assess its impact on other variables. For example: Kuttner (2001), Gürkaynak et al. (2005), Bernanke and Kuttner (2005), Hanson and Stein (2015), Bauer and Swanson (2022) estimate the effects of monetary policy on financial variables. Our study is (in terms of methodology) most similar to the studies of Jarociński and Karadi (2020) and Jarocinski (2020), which distinguish monetary shocks according to the high-frequency response of the stock market and Fed fund futures. As a result, the authors can distinguish between two different types of shocks, which have the same

impact on Fed funds futures, but a different impact on the stock market. Our approach is very similar, although we do not use high-frequency data. Instead, we use the revisions of expectations regarding the output and the price level of individual analysts, assuming that different types of shocks affecting a real economy cause different revisions in expectations. This is a similar approach to the one used in studies by Andrade et al. (2016), Carvalho et al. (2021).

Specifically, we use a Survey of Professional Forecasters to identify supply and demand shocks, which we identify using the revision of expectation in the price level and GDP/GNP. Our identification is generally straightforward: If there is an upward revision of expectations about GDP/GNP and inflation, we assume that a (positive) demand shock hits the economy in the given period. On the other hand, if there is a revision of the GDP/GNP and inflation, which is in the opposite directions, then we assume that supply shocks prevail in the economy. Given that we rely only on the differences of the expected values, these are rather to be considered perceived shocks as they do not necessarily have to coincide with the real economy. Also, as some recent research shows (Bauer and Swanson (2021), Steffensen et al. (2021)), central banks and financial markets do not possess significantly different information. Therefore, supply and demand shocks can be expected to be perceived similarly by the monetary authority, and we can also examine how monetary policy responds to these shocks by examining the movements in the short-end of the yield curve.

Concerning our identification strategy, a natural concern is that some of the movement in bond yields could be unrelated to supply and demand shocks and could instead reflect some type of yield shock. This is a valid argument, but we argue that it is a more likely explanation that markets generally respond to the current macroeconomic environment. The same applies to central banks, which are more likely to respond to current news rather than intentionally deviating from their monetary policy rules. Therefore, this paper does not consider these types of shocks. This assumption is in line with Bauer and Swanson (2021), which shows that publicly available news causes the private sector to revise its forecasts and the Fed to change monetary policy. To conclude, the bias in our estimation due to the existence of yield/monetary shocks depends on the size of these shocks. If these shocks explain only a small fraction of variation macroeconomic series, then the bias will be small. This is supported by the research from Faust et al. (2004), or Carvalho et al. (2021).

2. Data

We use US data from the Survey of Professional Forecasters (SPF) and additional data regarding Fed funds rate (from FRED) and bond yields. We use zero-coupon Treasury yields from Gürkaynak et al. (2007). We include maturities in the range of 1-15 years. Regarding SPF data, we use aggregated mean value (growth rate) data for the following series: GDP/GNP and GDP/GNP price deflator. Note that research by Mboup and Wurtzel (2018) shows that there are basically no differences between the mean and median forecasts for real GDP and inflation. We use all available data, i.e., the period from 1969 (q4) to 2022 (q1). Regarding frequency, the data are available only on a quarterly frequency. To determine revisions of expectations, we use the difference between the expected value for the quarter in which the Survey is conducted, and we compare this value with the expectations for this quarter in the previous round of the Survey, i.e., $E_t x_t - E_{t-1} x_t$, where x_t is a (forecast of) either GDP/GNP (y_t) or inflation (π_t). The Survey sets the deadlines for responses in the second to the third week of the middle month of each quarter when the Survey is conducted. Therefore, we use the end-of-month data for other series. Due to the importance of expectations, we use data for these nearest forecast periods, which should be the least sensitive to changes in expectations.

Regarding yield curve analysis, we use multiple representations of the yields. As a starting point, we include the 10-year yield together with the Fed funds rate. To generalize the results for the whole yield curve, we replace the single yield² by the level, slope, and curvature of the yield curve using Nelson and Siegel (1987) methodology:

$$y_t(\tau) = L_t + S_t \left(\frac{1 - e^{\lambda\tau}}{\lambda\tau} \right) + C_t \left(\frac{1 - e^{\lambda\tau}}{\lambda\tau} - e^{\lambda\tau} \right) \quad (1)$$

$y_t(\tau)$ denotes the yield with time to maturity τ . L_t , S_t and C_t denote the level, slope and curvature of the yield curve, λ is a parameter fixed to the value of 0.0609 (Diebold and Li, 2006). Table 1 presents a summary of the data used:

²We omit the Fed funds rate. The slope factor reflects the fluctuations on the short end of the yield curve and should provide sufficient information regarding the short rate movements.

Table 1: Data Description

Name (abbreviated)	Ticker	Source
Federal Funds Effective Rate	FEDFUNDS	FRED
GDP/GNP price index	PGDP	SPF
Real GDP/GNP	RGDP	SPF
The U.S. Treasury Yield Curve: 1961 to the Present	SVENYXX	federalreserve.gov
Name (abbreviated)	Mean	Standard Dev.
GDP/GNP price index revision	0,01	0,66
Real GDP/GNP revision	-0,30	1,70
Federal Funds Rate	4,98	3,90
10y gov bond yield	6,10	3,15
Level	6,70	2,76
Slope	-1,62	1,90
Curvature	-2,51	3,09

Note: The upper part of the table shows the data sources used. The lower part of the table shows the summary statistics of the actual data used in the estimation.

Note: Left panel shows GDP/GNP Deflator revisions. Right panel shows GDP/GNP revisions. GDP revision was winsorized (1 observation) for the graph purpose. Shaded regions denote NBER-dated recessions.

3. Methodology

In terms of methodology, we generally follow Jarociński and Karadi (2020).³ We use structural VAR with signs restriction to identify the impact of supply and demand shocks. More specifically, we use sign restrictions in conjunction with revisions of expectations to identify these shocks. Note that this decomposition is generally not unique but allows supply and demand shocks to hit the economy simultaneously. We use Minnesota-style priors as in Jarociński and Karadi (2020) to prevent the overfitting of the model. However, the estimation does not differ much from the frequentist version. Specifically, we take the prior for all coefficients of each variable to be zero, and we use standard Minnesota prior for their variance. We set the tightness parameter equal to 5 and decay parameter equal to 1, i.e., $\lambda_1 = 5$, $\lambda_2 = 1$. Furthermore, for the VAR covariance matrix, we use Inverse Wishart prior $IW(S, v)$, where we set $v = N + 2$, where N denotes the number of series used, and S is a diagonal matrix with the standard errors. We also impose block exogeneity on the revision of expectations, which means that the forecast revisions are unpredictable, i.e., $A_t, B_t = 0$.⁴ Equation 2 represents the VAR model:

$$\begin{pmatrix} REV_t \\ X_t \end{pmatrix} = \sum_{n=1}^p \begin{pmatrix} A_t & B_t \\ C_t & D_t \end{pmatrix} \begin{pmatrix} REV_{t-n} \\ X_{t-n} \end{pmatrix} + \begin{pmatrix} 0 \\ \alpha \end{pmatrix} + \begin{pmatrix} e_t^{rev} \\ e_t^X \end{pmatrix} \sim N(0, \Sigma) \quad (2)$$

REV_t is a vector of surprises which consist of revision of GDP and inflation, i.e.

³We only slightly modified their (publicly available) original code.

⁴The results don't differ much if we relax this assumption.

$E_t x_t - E_{t-1} x_t$, where $x_t = (\pi_t, y_t)$. X_t is a vector of variables of interest which consist either of Fed funds rate, and 10-year yield or yield curve factors (slope, level, curvature). We use two lags ($p = 2$), but the results are generally robust to any number of lags (1-4). We further impose the zero and sign restrictions but only to the revisions of expectations to identify supply and demand shocks, i.e. $shock_{supply} = (E_t \pi_t - E_{t-1} \pi_t) > 0 \wedge E_t y_t - E_{t-1} y_t < 0 \wedge X_t = 0$ and $shock_{demand} = (E_t \pi_t - E_{t-1} \pi_t) > 0 \wedge E_t y_t - E_{t-1} y_t > 0 \wedge X_t = 0$. Other series (X_t) remain unrestricted.⁵

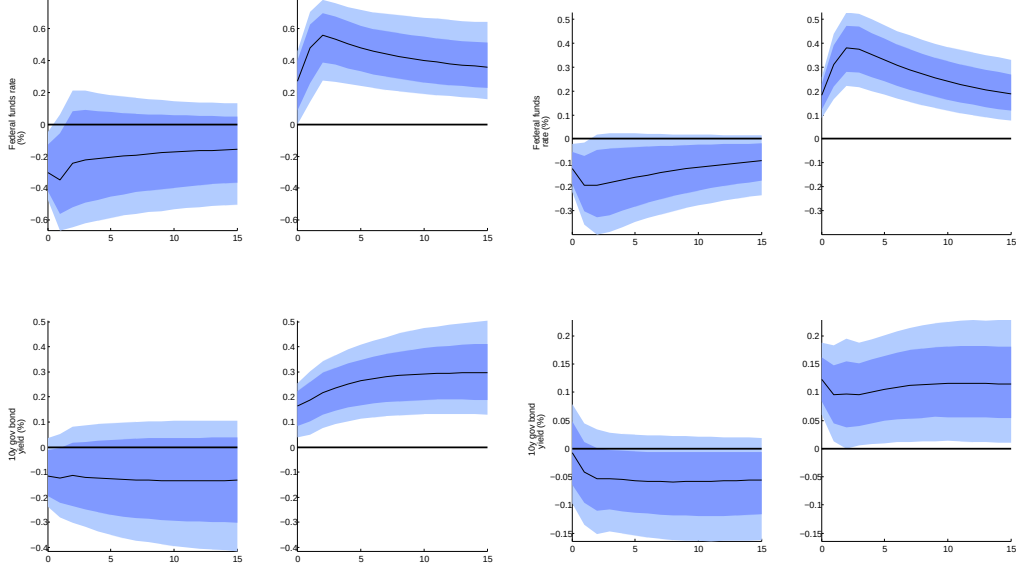
4. Empirical Results

Figure 1 left panel shows the model's results using full-sample data. A (negative) supply shock is associated with a 0,415 % increase in the revision of the inflation forecast and a 1,1578 % drop in the revision of the output forecast. This is due to the model construction, where these responses have to satisfy the sign restrictions. In the case of the supply shock, the responses are not clearly identified, but there generally seems to be some downward pressure on the Fed funds rate and the 10-year yield. Specifically, in the case of the Fed funds rate, the downward pressure is equal to 0,2 % for all quarters. In the latter case, the downward movement is 0,1 %. The impact of demand shocks on monetary policy and yields is generally more clearly identified. A demand shock is associated with a 0,485 % increase in the revision of the inflation forecast and a 1,28 % increase in the revision of the output forecast. The model confirms the usual institution that the rise of inflation (due to the demand shock) causes the rise of the Fed funds rate, which peaks in the first year and then slowly fades away. The 10-year yield also rises, but the increase is approximately half of the previous case. However, this increase seems to be longer-lasting.

Figure 1 right panel shows the results of the robustness check, which uses only a sub-sample of data. The sub-sample covers the period from 1988 (q1) to 2022 (q1). This is motivated by the desire to cover a more homogeneous period characterized by predominantly stable inflation. The general finding is that the impact of supply/demand shocks is similar to full-sample estimation. Our finding shows that a (positive) demand shock, associated with a 0,197 % increase in the revision of the inflation forecast and a 0,904 % increase in the revision of the output forecast, causes a rise in the Fed fund rates. The 10-year yield rises but significantly less than the Fed funds. In the case of the (negative) supply shock, associated with a 0,329 % increase in the revision of the inflation forecast and a 1,151 % drop in the revision of the output forecast, there seems to be some downward pressure. Contrary to the full-sample, there seems to be less significant downward movement in bond yields. Generally, the shock impacts (and the shocks) seem less significant in size than in the full-sample estimation.

⁵We are aware of the criticism of VAR with signs restriction by Baumeister and Hamilton (2020) about the Haar prior (un)informativeness for the structural impulse responses (as well as the response by Inoue and Kilian (2021)). Alternatively, we could directly divide supply and demand shock based on the sign restrictions: $shock_{supply(demand),t} = (E_t \pi_t - E_{t-1} \pi_t)$ if $(E_t y_t - E_{t-1} y_t) * (E_t \pi_t - E_{t-1} \pi_t) < (>) 0$, else 0. And then use these shocks as instruments.

Figure 1: Impulse Responses - Fed funds rate, 10-year yield
 (A) Full sample - 1969-2022 (B) Sub-sample - 1988-2022
 Supply shock Demand shock Supply shock Demand shock



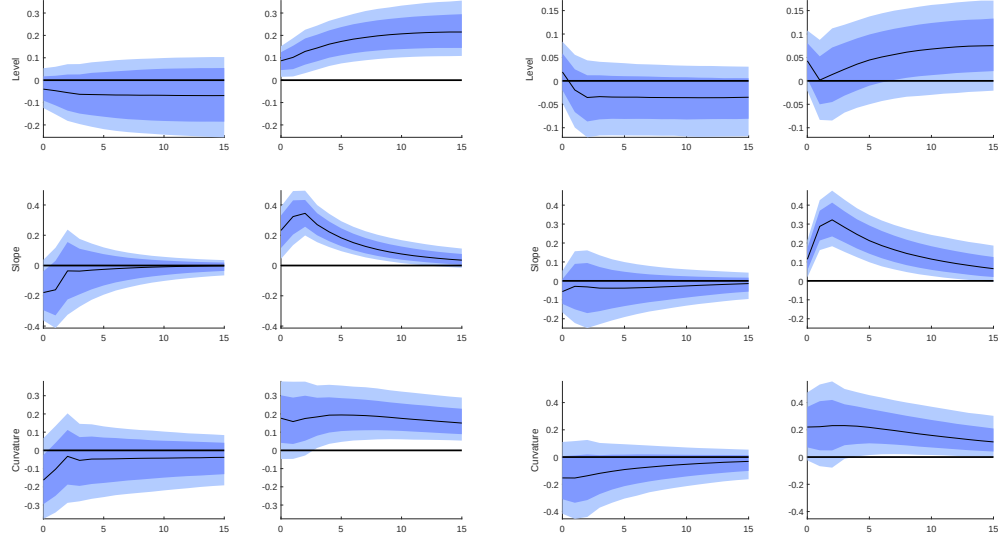
Note: Median (line), percentiles 16-84 (darker band), percentiles 5-95 (lighter band).

Figure 2 left panel shows the impulse-responses using Nelson-Siegel decomposition. In the case of the supply shock, associated with a 0,44 % increase in the revision of the inflation forecast and a 1,171 % drop in the revision of the output forecast, the yield curve response is not clearly identified. However, there seems to be some downward pressure on the level. Specifically, the downward pressure is equal to approximately 0,05 % for all quarters. On the other hand, supply shock has only a short-lived effect on the slope and curvature. This means that supply shock causes a parallel shift in the yield curve. A demand shock is associated with a 0,485 % increase in the revision of the inflation forecast and a 1,273 % increase in the revision of the output forecast. The results show that demand shock is associated with the rise of the level, which seems to be long-lasting. Also, slope and curvature rise, this increases short and medium maturities more than long yields, but this effect slowly fades away. Specifically, slope response peaks in the first year and then returns to zero.

Figure 2 right panel shows the results using only a sub-sample of data. This generally

Figure 2: Impulse Responses - Yield Curve

(A) Full sample - 1969-2022 (B) Sub-sample - 1988-2022
 Supply shock Demand shock Supply shock Demand shock



Note: Median (line), percentiles 16–84 (darker band), percentiles 5–95 (lighter band).

confirms the results of the full-sample estimation. A negative supply shock is associated with a 0,198 % increase in the revision of the inflation forecast and a 0,89 % drop in the revision of the output forecast. Contrary to the full-sample, there seems to be even less significant downward movement in level and slope. Interestingly, the main difference is regarding curvature, with a more negative and longer-lasting response. Positive demand shock is associated with a 0,33 % increase in the revision of the inflation forecast and a 1,51 % increase in the revision of the output forecast; it causes a rise in all the factors, and results generally do not differ much from the full-sample estimation. The main difference is a less-significant response of the level.

5. Conclusion

In this paper, we present a simple way of extracting the supply and demand shock using the revisions of expectations of analysts using the Survey of professional forecasters. The main contribution is the identification of these shocks and the subsequent assessment of their impact on Fed funds rate and bond yields. Specifically, we use structural VAR with signs restrictions to assess the impact of these shocks. The study empirically shows the different impacts of supply and demand shocks on bond yields. Overall, the study allows for a better structural interpretation of yield curve dynamics.

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